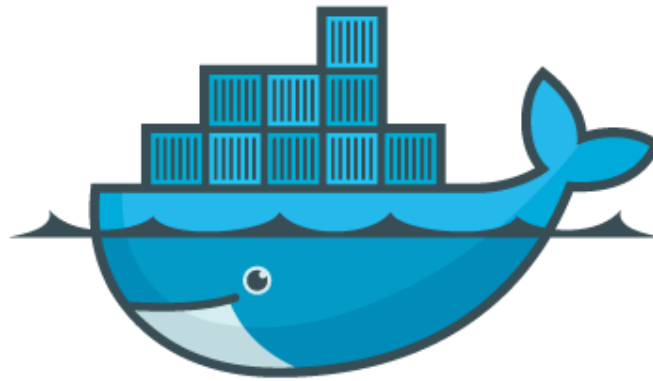


DOCKER AND JBOSS

THE PERFECT COMBINATION

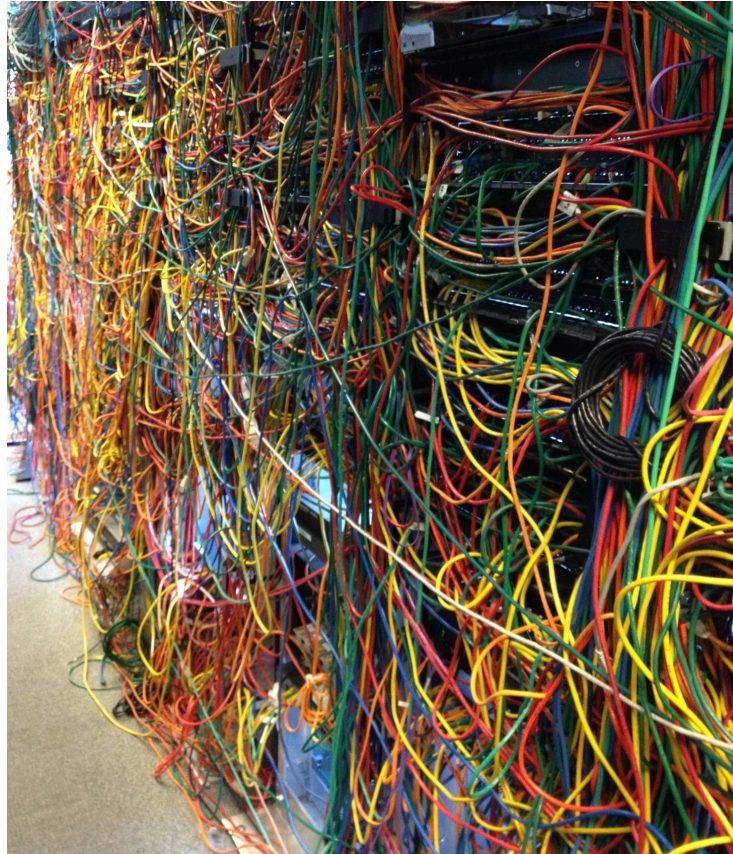
@marekgoldmann

Toruń JUG, 29th Oct 2014



docker

APPLICATIONS ARE COMPLEX

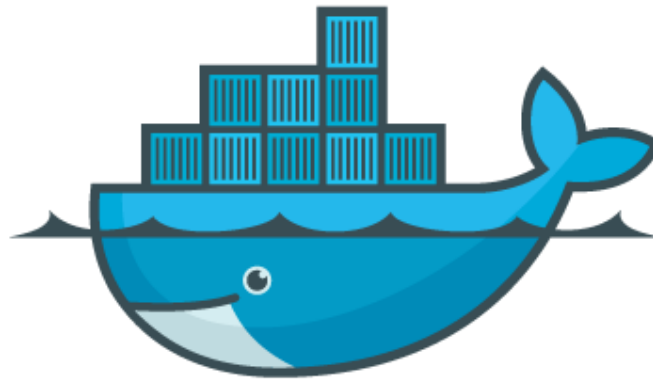


http://www.reddit.com/r/pics/comments/nf8la/im_always_tempted_to_switch_two_random_cables_to/

DIFFERENT DEPLOYMENT ENVIRONMENTS



<http://www.shedworking.co.uk/2012/03/nuts-and-bolts-cable-ties.html>



docker

A project to manage containers.

CONTAINERS?

Yes, a lightweight operating system virtualization.

HOW THIS COMPARES TO A VIRTUAL MACHINE?

It's **completely** different.

COMPARISON

	Virtual Machine (KVM, VMware)	Container (LXC, OpenVZ)
Hardware	Simulated	Uses it (almost) directly
Supported OS'es	Almost any	Only Linux
Space	User space	Kernel space
Separation	Full	Control Groups (cgroups)
Startup time	Seconds to minutes	Miliseconds
Scalability	A few	Sky is the limit (thousands)
Custom kernel	Yes	No
Enterprise features (live migration, etc)	Yes	No
Ease of creation	Moderate	Easy
Time consumption of creation	High	Low
Size	HUGE	Small

LINUX FTW!
(STILL)



Source: <http://goo.gl/bqb360>

<http://blog.docker.com/2014/10/docker-microsoft-partner-distributed-applications/>

[**https://github.com/boot2docker/boot2docker**](https://github.com/boot2docker/boot2docker)

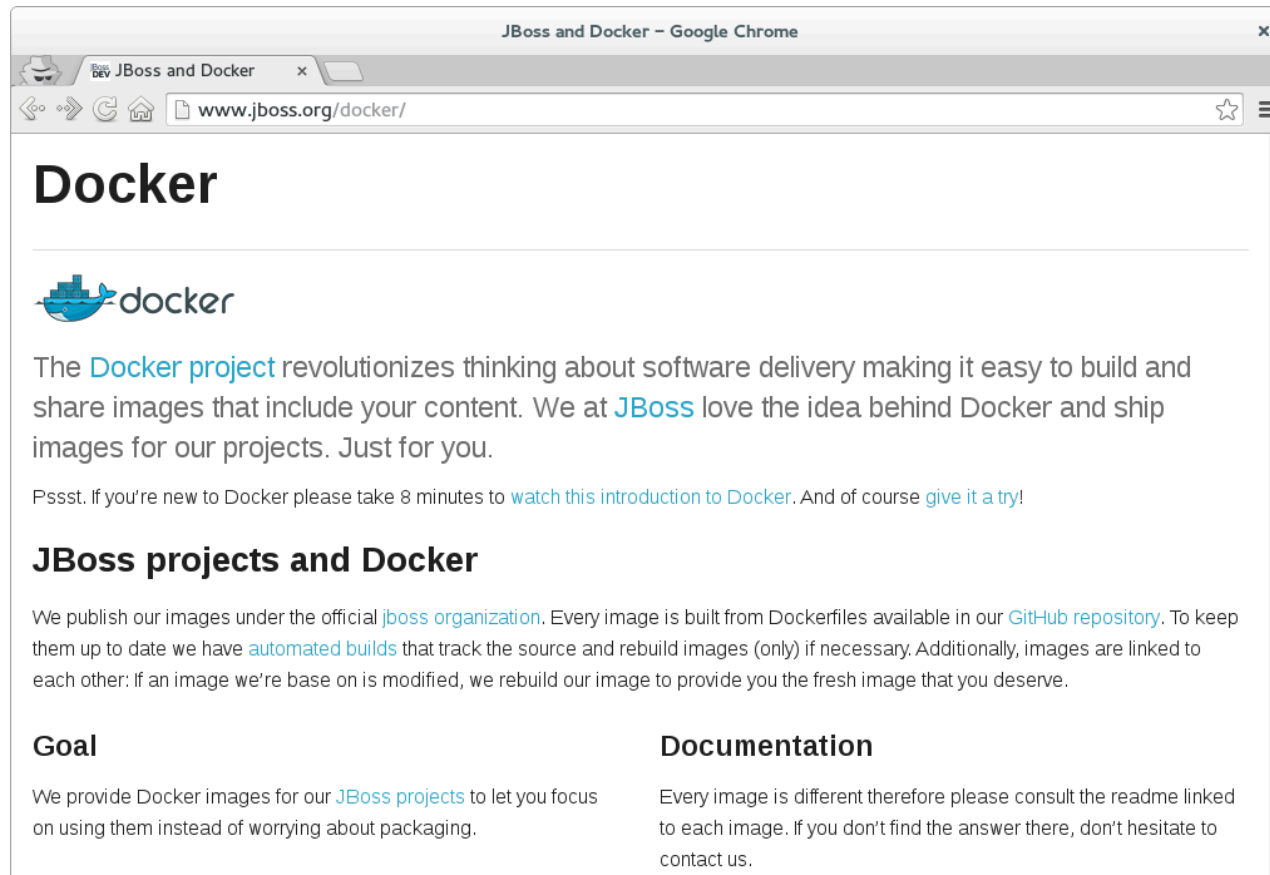
DOCKER COMMUNITY...

- **GitHub**
 - >1100 watchers
 - >14000 stars
 - >2700 forks
 - >660 open issues
 - >100 pull requests
- Over **585 contributors**
 - avg 17 commits / day
 - **90% are external**
- Many meetups all over the world
 - **50** active **meetup** groups

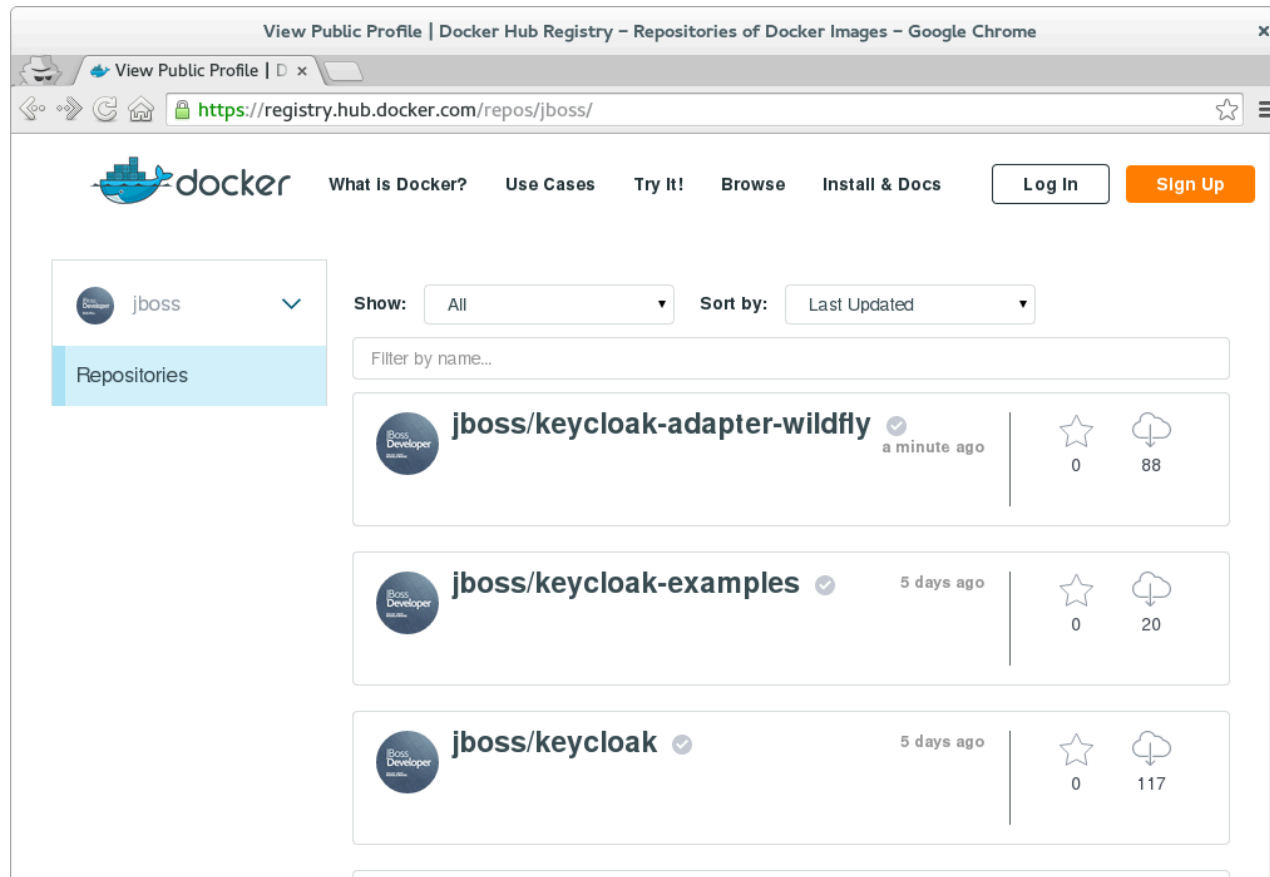
JBOSS AND DOCKER

DOCKERIZED PROJECTS

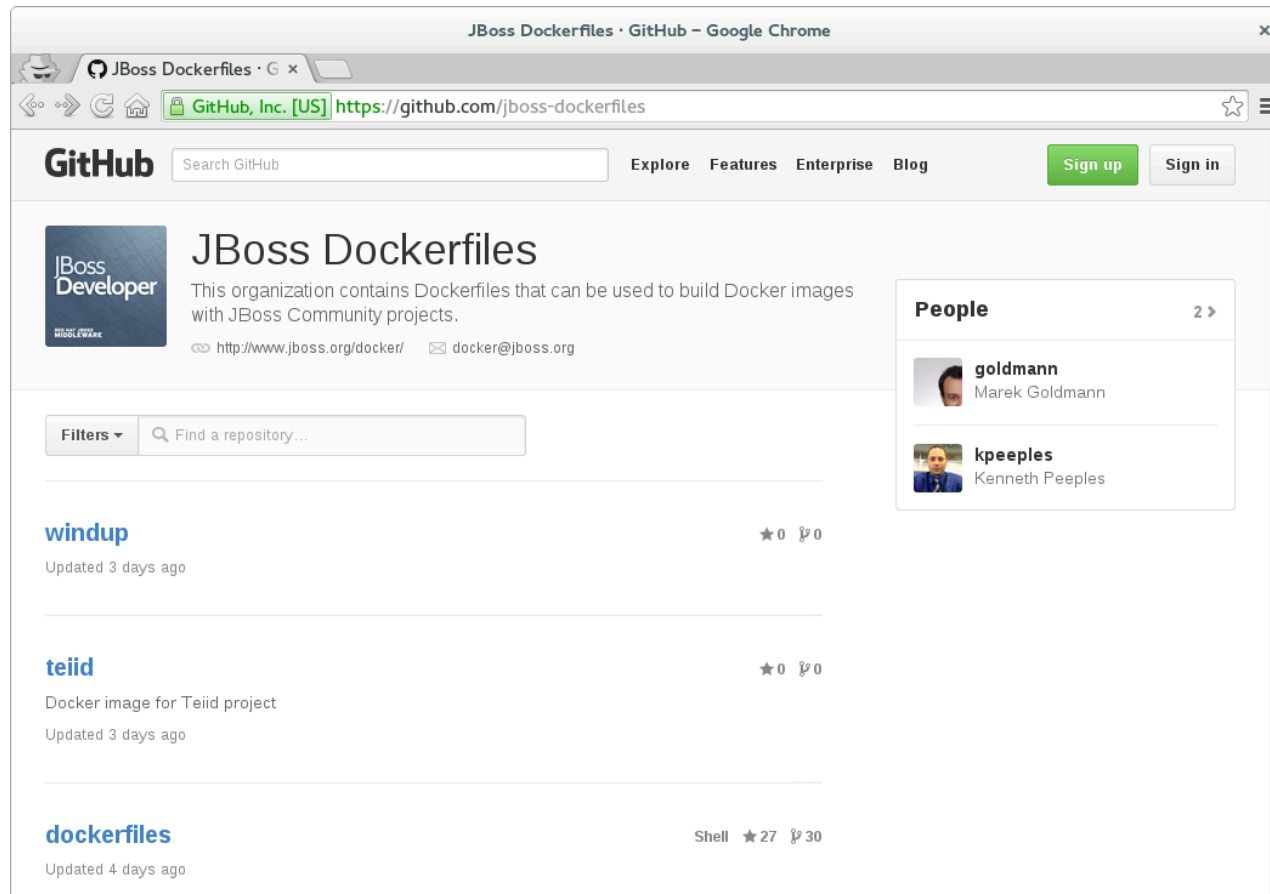
- WildFly
- Keycloak
- TorqueBox
- Aerogear
- Immutable
- Nodyn
- LiveOak
- Infinispan
- ModeShape
- ...



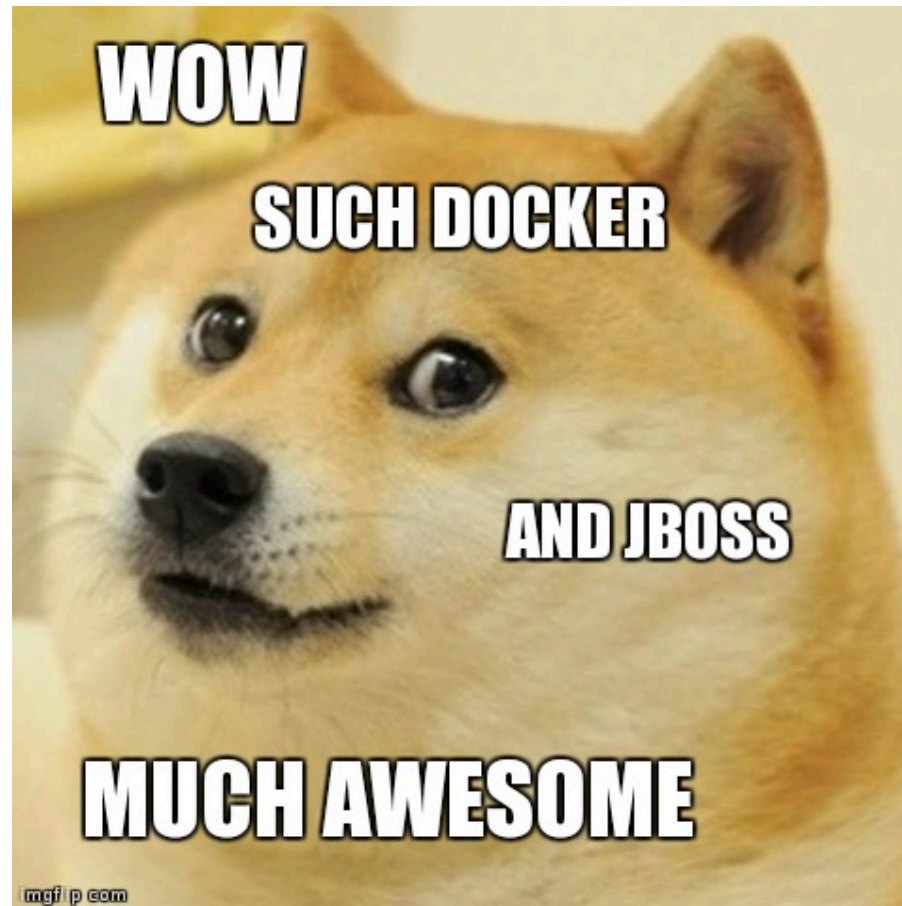
<http://www.jboss.org/docker/>



<https://registry.hub.docker.com/repos/jboss/>



<https://github.com/jboss-dockerfiles>



DEMOS!

A FEW TIPS

I NEED METRICS!

- Use iptables for network
- Parse /sys/fs/cgroup/ for everything else

I NEED TO LIMIT RESOURCES!

- CPU: `docker run -c 512 [...]`
- Memory: `docker run -m 128m [...]`
- Disk space: 10GB by default
 - **`docker -d --storage-opt dm.basesize=20G`**

I NEED TO DO BACKUP!

- Files?
 - Use volumes
- Data?
 - Use links

I NEED TO ACCESS MY CONTAINERS!

- Do you own the host?
 - Use nsenter
- In other cases:
 - Run sshd inside of the container (boooo...)

I NEED PERFORMANCE! RLY?

- CPU: native
- Memory: (almost) no overhead
- Network: no overhead with `--net host`
- I/O: native on volumes

THANKS!



Attributions:

<https://www.docker.io/>

